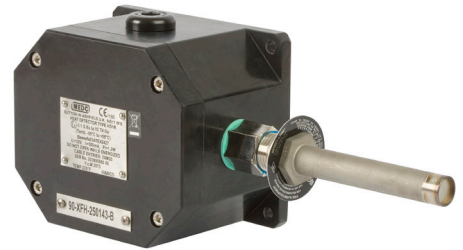


# HD1 heat detector range

## Hazardous locations



## Overview

The MEDC heat detector has been designed for use in hazardous environments.

These units are suitable for fire alarm and/or suppression systems in offshore and onshore applications including paint spray booths, flammable material stores, turbine rooms, extract ductwork and other hazardous areas throughout the oil & gas, petrochemical and process industries.

Comprising a Fenwal rate-compensated detector with all-stainless steel external construction, mounted to a JB10 corrosion-free GRP enclosure. The contact in the detector closes at alarm temperature.

To select appropriate temperature setting see specification on reverse.

## Features

- UL Listed for USA and Canada:  
Class I, Div. 2,  
Groups A, B, C, D
- ATEX certified
- IP66 & IP67
- Certified temperature:  
–4°F to +131°F  
–20°C to +55°C
- Stainless steel probe
- Detector temperature settings  
140°F to 725°F (60°C to 385°C)
- GRP enclosure
- Optional stainless steel guard



| Certifications     |                                                                                                                                                                          |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UL                 | Listing no. E252920 – versions up to 450°F.<br>Listing no. E254077 – versions from 600°F to 725°F.<br>UL for USA and Canada, listed to Class 1, Div 2. Groups A, B, C, D |
| SIL                | SIL2 certified. Cert no. Sira FSP 12007/02                                                                                                                               |
| Specifications     |                                                                                                                                                                          |
| Material           | Detector: stainless steel<br>Enclosure: GRP (anti-static)<br>Cover screws: stainless steel<br>Optional guard: 316 stainless steel                                        |
| Finish             | Detector: sand blasted<br>Enclosure: natural black or painted to customer's specification                                                                                |
| Weight             | 2.4lbs (1.1kg)                                                                                                                                                           |
| Certified temp     | -4°F to +131°F<br>-20°C to +55°C                                                                                                                                         |
| Ingress protection | NEMA 4X & 6, IP66 & IP67                                                                                                                                                 |
| Operation          | The detector contact is normally open and closes at alarm temperature                                                                                                    |
| Contact rating     | 125Vac - 5A, 125Vdc - 0.5A, 48Vdc - 1A, 24Vdc - 2A                                                                                                                       |
| Entries            | 2 x 1/2" NPT (via supplied adaptors)                                                                                                                                     |
| Terminals          | 6 x 12 AWG                                                                                                                                                               |
| Labels             | Optional stainless steel tag and duty labels                                                                                                                             |

**Listed temperature:** To select appropriate temperature settings, choose detector at settings: 100°F (38°C) above maximum ambient temperature

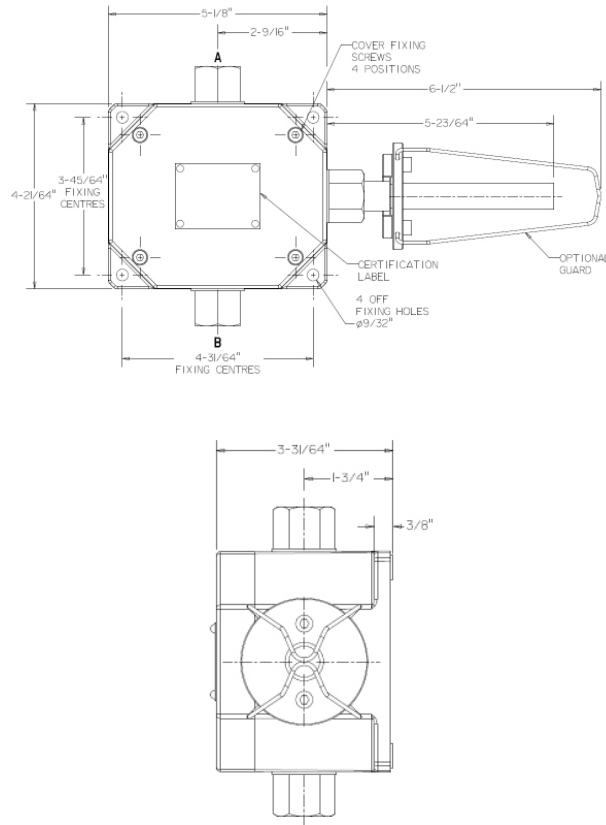
| Temperature setting |      | Tolerance |      | Color code of text on probe |
|---------------------|------|-----------|------|-----------------------------|
| (°F)                | (°C) | (°F)      | (°C) |                             |
| 140                 | 60   | +7/-8     | ±4   | Black                       |
| 160                 | 71   | +7/-8     | ±4   | Black                       |
| 190                 | 88   | +7/-8     | ±4   | White                       |
| 210                 | 99   | +7/-8     | ±4   | White                       |
| 225                 | 107  | +7/-8     | ±4   | White                       |
| 275                 | 135  | ±10       | ±6   | Blue                        |
| 325                 | 163  | ±10       | ±6   | Red                         |
| 360                 | 182  | ±10       | ±6   | Red                         |
| 450                 | 232  | ±15       | ±8   | Green                       |
| 500                 | 260  | ±15       | ±8   | Orange                      |
| 600                 | 316  | ±20       | ±11  | Orange                      |
| 725                 | 385  | ±25       | ±14  | Orange                      |

## Ordering requirements

The following code is designed to help in selection of the correct unit. Build up the reference number by inserting the code for each component into the appropriate box

| Model         | Certification | Temp settings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Options | Finish |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
|---------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|------|---|------------|----|-----------|----|-------|---|---------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|---------------|---|------|---|-----|---|------|---|--------|---|---------|----|
| HD1           | ULU           | <table border="1"> <thead> <tr> <th>Temp °F</th> <th>Code</th> </tr> </thead> <tbody> <tr><td>140</td><td>140</td></tr> <tr><td>160</td><td>160</td></tr> <tr><td>190</td><td>190</td></tr> <tr><td>210</td><td>210</td></tr> <tr><td>225</td><td>225</td></tr> <tr><td>275</td><td>275</td></tr> <tr><td>325</td><td>325</td></tr> <tr><td>360</td><td>360</td></tr> <tr><td>450</td><td>450</td></tr> <tr><td>500</td><td>500</td></tr> <tr><td>600</td><td>600</td></tr> <tr><td>725</td><td>725</td></tr> </tbody> </table> | Temp °F | Code   | 140 | 140 | 160 | 160 | 190 | 190 | 210 | 210 | 225 | 225 | 275 | 275 | 325 | 325 | 360 | 360 | 450 | 450 | 500 | 500 | 600 | 600 | 725 | 725 | <table border="1"> <thead> <tr> <th>Option</th> <th>Code</th> </tr> </thead> <tbody> <tr><td>None</td><td>N</td></tr> <tr><td>Duty label</td><td>D*</td></tr> <tr><td>Tag label</td><td>T*</td></tr> <tr><td>Guard</td><td>G</td></tr> <tr><td>Blanking plug</td><td>P</td></tr> </tbody> </table> <p>*Please specify</p> | Option | Code | None | N | Duty label | D* | Tag label | T* | Guard | G | Blanking plug | P | <table border="1"> <thead> <tr> <th>Finish</th> <th>Code</th> </tr> </thead> <tbody> <tr><td>Natural black</td><td>N</td></tr> <tr><td>Grey</td><td>G</td></tr> <tr><td>Red</td><td>R</td></tr> <tr><td>Blue</td><td>B</td></tr> <tr><td>Yellow</td><td>Y</td></tr> <tr><td>Special</td><td>S*</td></tr> </tbody> </table> <p>*Please specify</p> | Finish | Code | Natural black | N | Grey | G | Red | R | Blue | B | Yellow | Y | Special | S* |
| Temp °F       | Code          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| 140           | 140           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| 160           | 160           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| 190           | 190           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| 210           | 210           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| 225           | 225           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| 275           | 275           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| 325           | 325           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| 360           | 360           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| 450           | 450           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| 500           | 500           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| 600           | 600           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| 725           | 725           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| Option        | Code          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| None          | N             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| Duty label    | D*            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| Tag label     | T*            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| Guard         | G             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| Blanking plug | P             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| Finish        | Code          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| Natural black | N             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| Grey          | G             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| Red           | R             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| Blue          | B             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| Yellow        | Y             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |
| Special       | S*            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |                                                                                                                                                                                                                                                                                                                           |        |      |      |   |            |    |           |    |       |   |               |   |                                                                                                                                                                                                                                                                                                                                                   |        |      |               |   |      |   |     |   |      |   |        |   |         |    |

## General arrangement drawing (all dimensions in inches)



To order ATEX approved version, see European data sheet